# A MORPHISM CARRYING A RELATIVELY AMPLE INVERTIBLE SHEAF IS SEPARATED

THE STACKS PROJECT AUTHORS

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*Allowed standard prerequisites (editorial).*

- Separatedness is local on the base. Schemes, Lemma \ref{schemes-lemma-characterize-separated}, cited at proof lines 8664-8665. Reduces to an affine base.
- A scheme with an ample invertible sheaf over an affine base is separated. Properties, Lemma \ref{properties-lemma-ample-separated}, cited at proof lines 8668-8671. Gives the reduced statement.

*Source reference.* Original lemma 01VI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2447–2447. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 01KP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2058–2058. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

*Source reference.* Original lemma 09MP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12187–12187. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $f : X \to S$ be a morphism of schemes. If there exists an $f$-ample invertible sheaf, then $f$ is separated.*

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*Date*: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* Being separated is local on the base (see Schemes, Lemma 01KP for example; it also follows easily from the definition). Hence we may assume $S$ is affine and $X$ has an ample invertible sheaf. In this case the result follows from Properties, Lemma 09MP. □